DEPARTMENT OF CSE
CS T51 OPERATING SYSTEMS
Unit-I

Introduction: Mainframe Systems – Desktop Systems – Multiprocessor Systems –Distributed Systems – Clustered Systems - Real Time Systems – Hardware Protection –System Components – Handheld Systems - Operating System Services – System Calls –System Programs – Process Concept – Process Scheduling – Operations on Processes –Cooperating Processes – Inter-process Communication.

PART A

1. **Define the term 'operating systems' with examples. (NOV '14)**
2. What is the duty of a command interpreter?
3. What is meant by loosely coupling?
4. What are the goals of operating system?
5. What is multitasking?
6. List the process states.
7. What are the basic components of a computer?
8. Draft a note on system calls.
9. What is meant by multiprogramming?
10. State the function of system calls.
11. **Mention the categories of system programs. (NOV '14)**
12. Define spooling.
13. Define process.
14. Define message passing.
15. What is meant by graceful degradation?
16. **What is batch processing system? (APR '15)**
17. Draft a note on handheld system.
18. Depict with a neat sketch the process involved in hardware protection.
19. What are the four different types of operating system structures?
20. What is a JVM?
21. Define Booting.

22. Define Kernel.
23. Draft a note on RAM & ROM
24. Write about time sharing systems.
25. **Define Process.(Nov 2012)**
26. Define process state.**(Nov 2014)**
27. What is scheduler?**(Nov/Dec2013)**
28. What is meant by Dispatching?**(Nov 2012)**
29. What is PCB?
30. List the process state.
31. Define IPC.
32. **Define Context switch**

PART B

1. Give a detailed note on multiprocessor, distributed and clustered systems.(NOV'14) (APR '15)
2. Explain the system calls with examples. (APR '15)
3. Explain any five operating system services. (APR '13)
4. Discuss cooperating processes in detail.
5. Describe the categories of system programs.
6. Describe Process scheduling in detail.
7. Discuss in detail IPC. (APR '13)
8. Draft a detailed note on operation process.

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Threads: Overview – Threading issues - CPU Scheduling – Basic Concepts – Scheduling Criteria – Scheduling Algorithms – Multiple-Processor Scheduling – Real Time Scheduling -The Critical- Section Problem – Synchronization Hardware – Semaphores – Classic problems of Synchronization – Critical regions – Monitors.

PART A (2 MARKS)

1. Define Thread.
2. Depict a neat sketch of LWP.
3. Draft a note on CPU Scheduler.
4. Explain in brief Dispatcher.
5. List out the CPU Scheduling criteria.
6. Give a brief note on Real time scheduling.
7. **State any two classic problems of synchronization.(APR ’15)**
8. **Define turnaround time and response time.(APR ’13)**
9. **List the two types of threads.(DEC ’14)**
10. **Draft a note on preemptive and non-preemptive scheduling.(DEC ’14)**
11. Define entry section and exit section.
12. What are the use of job queues, ready queues & device queues?
13. **What is meant by CPU Scheduling?(NOV 13)**
14. What are the types of Scheduling available?
15. What is meant by Priority Scheduling?
16. What is Preemptive Scheduling?
17. What is Non - Preemptive Scheduling?
18. What are the properties of Scheduling Algorithms?
19. What is meant by First Come, First Served Scheduling?
20. What is meant by Shortest Job First Scheduling?
21. Define critical section problem.
22. Define semaphore.
23. Draft a note on monitors.

24. Define mutual exclusion.
25. Draft a note on Bounded Waiting.
26. Explain in brief critical region.
27. Draft a note on progress.
28. Define Round robin scheduling.
29. List the various CPU optimization criteria.
30. Depict the Alternating Sequence of CPU And I/O Bursts with a neat sketch

PART B (11 MARKS)

1. Give a detailed note on the concept of threads.
2. Explain in detail the basic concept of CPU scheduling.
3. Draft a detailed note the FCFS, SJF, PRIORITY and ROUND ROBIN CPU Scheduling algorithm**.(DEC ’14) (NOV ’13) (APR ’13)(APR ’15)**
4. Give a detailed note the critical section problem.**(DEC ’14)**
5. Draft a detailed note on monitors.
6. Explain in detail the various classical problems of synchronization**.(APR ’13)**
7. Give a detailed note on critical regions.
8. Explain in detail multilevel queue scheduling, multilevel feedback queue scheduling, and multiple processor scheduling**.(APR ’15)**
9. **Draft a detailed note on semaphore.(NOV ’13)**

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System Model: Deadlock Characterization – Methods for handling Deadlocks -Deadlock Prevention – Deadlock avoidance – Deadlock detection – Recovery from Deadlocks - Storage Management – Swapping – Contiguous Memory allocation – Paging – Segmentation – Segmentation with Paging.

PART A (2 MARKS)

1. What will be the sequence in which the process may utilize a resource under normal mode of operation?
2. **Define deadlock. (UQ NOV'13)**
3. **Enumerate the necessary conditions that lead to a deadlock. (APR'15)(NOV'14)**
4. Define mutual exclusion.
5. Draft a note on hold & wait, no preemption.
6. Define circular wait.
7. **Briefly explain system resource allocation graph. (UQ APR'12)**
8. Depict a neat sketch exhibiting Resource allocation graph with and without deadlock.
9. Give a brief note on request edge and assignment edge.
10. Enumerate and brief the methods for handling deadlocks.
11. Compare and comment on deadlock prevention & Avoidance.
12. Define safe state.
13. Depict with a neat sketch safe, unsafe and deadlock state space.
14. **Draft a brief note on banker's algorithm. (UQ APR'12)**
15. Define starvation
16. List out the steps involved in binding of instructions and data to memory address.
17. Depict with a neat sketch the multistep processing of a user program.
18. Define physical logical and virtual addresses.
19. Draft a note on MMU.
20. Define physical & logical address space.
21. Define dynamic loading.
22. Depict a neat sketch showing dynamic relocation using a relocation register.
23. Define overlays.

24. **Draft a note on swapping. (APR'15)**
25. Depict a neat sketch explaining the process of swapping.
26. **Draft a note on external and internal fragmentation. (APR 2014)**
27. Define compaction.
28. Define paging.
29. What is the purpose of frame & pages in the process of paging?
30. Brief a note on PTBR.
31. Define TLB.
32. Draft a note on ASID.
33. Define segmentation.
34. **Why the page sizes is always power of 2 in memory? (UQ NOV'13)**

PART B (11 MARKS)

1. a) Explain in detail the condition necessary for the occurrence of a deadlock. (5)

   b) Draft a detail note on RAG. (6)
2. **Give a detailed note on the ways to be adapted for preventing a deadlock. (APR'15)**
3. **Explain the importance of deadlock avoidance with available algorithm used to serve the purpose. (UQ APR'12)**
4. **a) Explain in detail how deadlock is detected with suitable illustrations.(6)**

   b) What are the ways by which recovery for deadlock is made? (5) **(UQ NOV'11& NOV'12)**
5. Explain in detail

   i) Dynamic Loading. (4)

   ii) Dynamic Linking & Shared Libraries. (4)

   iii) Overlays. (3)
6. Draft a detailed note on swapping.
7. **Explain Contiguous Memory Allocation in detail. (UQ APR'14)**
8. **Give a detailed note on paging. (UQ APR'14) (UQ APR'12)**
9. **Draft a detailed note on Segmentation. (UQ NOV'13)**

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Virtual Memory – Demand Paging – Process creation – Page Replacement – Allocation of frames – Thrashing - File Concept – Access Methods – Directory Structure – File System Mounting – File Sharing – Protection.

PART A (2 MARKS)

1. Define ***virtual memory***. **(APR '11)(APR '12)**
2. Draft a note on ***virtual address space***.
3. Draft the ***methods through which the virtual memory concept in implemented***.
4. Draft a note on ***demand paging***.
5. Draft a note on ***lazy swapper***. **(NOV '13)**
6. Enumerate the ***advantages of using a virtual memory***.
7. What do you mean by ***pure demand paging***?
8. Define ***locality of reference***.
9. Define ***Bélády's anomaly***.
10. List out the various page replacement algorithm.
11. Define ***thrashing***. **(APR 11)**
12. Draft a note on ***working set model***. **(NOV'14)**
13. Draft a note on search path. **(NOV '13)**
14. Define file.
15. Give a brief note on the various ***file attributes***. **(APR '12)(NOV '12)**
16. List out the various file operations.
17. Give a brief note on the various ***file types***.
18. Enumerate the various ***file access method***. **(APR'15)**
19. Enumerate the operations that are performed on a directory.
20. Draft a note on directory.
21. List the various logical structures of a directory.
22. Draft a note on ***UFD & MFD***.
23. Define ***file system mounting***.
24. Define ***consistency semantics***.

25. Justify! Why protection mechanism is very much important with respect to file system?

26. Define ***ACL***. **(NOV'14)**

27. List out the classification of users in connection with accessing a file.

28. What is ***System file checker in Windows***? **(NOV '11).**

29. What is ***File Lock***? (**NOV '11) (APR '11).**

30. What is ***File Control Block***? **(APR '14).**

31. What is ***File Authentication***?**(NOV '11)(ARPIL '11)**

PART B (11 MARKS)

1. Give a detailed note on the need and implementation strategy of virtual memory.
2. Draft a detailed note on demand paging.
3. Explain in detail process creation.

4. Elaborate the process of page replacement with neat diagram. **(APR'15)**

5. Explain the various page replacement algorithms with suitable example.
6. a) Explain in detail allocation of frames.(6)

   b) Elaborate the concept of thrashing in detail. (5) **(APR '11) (APR '14)**

7. Draft a detailed note on the various file access methods. **(APR '14)**

8. Draft a detailed note on the various directory structures with neat sketch. **(APR '11) (NOV 13)**

9. a) Explain file sharing methods in detail.(6) **(APR '12)**

b) Comment on the importance of implementing protection mechanism with respect to files. (5) **(APR'15)**

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File System Structure – File System Implementation – Directory Implementation – Allocation Methods – Free-space Management. Kernel I/O Subsystems - Disk Structure – Disk Scheduling – Disk Management – Swap-Space Management. **Case Study**: The Linux System, Windows.

PART A (2 MARKS)

1. **Define Bootstrap Program?(NOV '12)**
2. What is boot control block?
3. What is volume control block?
4. Depict file control block.
5. List the methods to implement directory
6. Write the types of directory allocation methods.
7. What is Contiguous Allocation?
8. What is the drawback of contiguous allocation? How to overcome from it?
9. What is linked allocation?
10. Write the disadvantages of Linked allocation.
11. What is indexed allocation?
12. Write about free space management.
13. List the strategies by which free list can be implemented.
14. Write about Bit vector.
15. Write about Linked List in free space management.
16. Write about Grouping in free space management.
17. Write about Counting in free space management.
18. **What do you mean by seek time? (APR '12)**
19. **List the disk performance parameter.(APR '13)**
20. **What is the purpose of device drivers? (APR '13)**
21. List Different types of scheduling algorithms.
22. Define Slack ware.
23. **Define Kernel in OS. (APR '12) (APR'15)**

24. Define Module Management.
25. What operations are available in process management?
26. Define Scheduling Context.
27. Define signal-Handler table.
28. **What is IPC? (APR ‘12)**
29. **Write any two advantages of Linux?(NOV ‘12)**
30. Write any two design principles of Linux?
31. Does Linux support Symmetric Multiprocessing?
32. **List the role of process manager.(NOV ‘13)**
33. **Brief about Process Environment Block (PEB).(NOV ‘13)**
34. Write about fork() and clone() system call?
35. Write about few design principles of windows XP
36. **What type of security is provided by Windows XP (APR’15)**
37. Write about system components of XP?
38. **Sketch the components of a Linux system (NOV’14)**

PART B -11 MARKS

1. Explain the allocation methods for disk space in file allocation? (NOV 12, 13) (APR’11, 15).
2. Explain Disk Scheduling in detail with example? (APR ‘11) (APR ‘13).
3. Briefly explain Disk management and swap space management? (NOV ‘12).
4. Explain in detail about Kernel architecture of Linux? (APR ‘13)(NOV ‘13).
5. Explain in Detail about Process management? (NOV’14).
6. Explain how inter process communication is carried out in Linux OS? (NOV‘12)(APR’13,15).
7. Explain how the memory management is implemented in windows XP? (APR’13)
8. Explain in detail about Windows XP System Architecture?(NOV ‘12) (APR’15)
9. Explain how the following mechanisms are implemented in Windows XP? (1)Thread (2) IPC (APR ‘12).
10. Differentiate between Windows and Linux operating system. (NOV 14).
11. Explicate about Windows Security.